

# Samuel Xavier Borges

contactme@samborg.dev | linkedin.com/in/samuel-xavier-borges | github.com/samborg-dev

## EDUCATION

### University of Central Florida

2028

Bachelor of Science in Computer Science, Minor in Data Science, Minor in Studio Art | GPA: 3.5

Orlando, FL

## EXPERIENCE

### Undergraduate Research Assistant

May 2026 – Present

Unary Lab - University of Central Florida

Orlando, FL

- Refactoring a **Python/PyTorch** research codebase that profiles transformer nonlinear activations (**softmax**, **GELU**, **SiLU**) and benchmarks hardware-oriented approximations on **Hugging Face** models.
- Improving experiment reproducibility by modularizing modality-specific inference pipelines, centralizing **YAML-driven** configuration, and streamlining workflows used to evaluate accuracy on **SLURM GPU** clusters.

### Software Engineering Intern

Feb 2025 – Jan 2026

Knight Hacks

Orlando, FL

- Improved site accessibility and maintainability by developing reusable **React UI** components to replace non-semantic SVG elements and embed proper **ARIA** attributes, enhancing keyboard navigation compliance.
- Developed a ground-up email automation system using the **Google API** to send personalized acceptance, denial, and event-detail communications to applicants for one of **Florida's largest hackathons**.

### Software Engineering Intern

May 2025 – Aug 2025

Data-Enabled Photovoltaics

Orlando, FL

- Automated parsing and normalization of heterogeneous instrument metadata by developing an **object-oriented Python** parser with **Pandas** and **SQLite**, reducing manual preprocessing time by over **75%**.
- Achieved consistent metadata provenance and context tracking by defining and validating **FAIR** data model standards and integrating workflow outputs across pipelines.

### Software Engineering Intern

Mar 2025 – May 2025

Zuleris Interactive

Orlando, FL

- Developed a modular **Unity/C# radio interference simulator** with configurable overlap frequencies and **coroutine-driven** randomized jamming, enabling a fully adjustable multi-radio training tool.
- Implemented global static **C#** events to broadcast **UI** open/close and frequency changes, synchronizing audio playback and tuning across **four** independent radio instances for seamless, consistent behavior.

## PROJECTS

### Ctrl-ARM | Python, Pandas, Electron, React, Google ADK, TensorFlow, Whisper

- Created a body movement **accessibility** system that combined **EMG** signals with **IMU** motion data and a multimodal **Electron + React** overlay, achieving **1st at ShellHacks 2025's Microsoft AI Challenge**.
- Engineered **Pandas ETL** pipelines to ingest and transform 200Hz EMG/IMU streams, enabling per-user calibration, feature extraction, and consistent **90%** classification accuracy across **12+** gestures.

### Tariffix | TypeScript, React, Vite, Tailwind CSS, MongoDB, Chrome Extension API, Gemini API, Cheerio

- Enabled real-time tariff insights on **100K+** product listings via a **Vite-powered Chrome extension** backed by an **AI scraping pipeline** for metadata and HTS codes, winning **Bitcamp 2025's "Best use of MongoDB."**
- Delivered tariff cost estimates against official U.S. rates with under **5%** average error by comparing **AI-scraped data** to a **Gemini-powered MongoDB dataset**, automating precise calculations for end users.

### Kmodo | TypeScript, Next.js, React, tRPC, Tailwind CSS, Drizzle, PostgreSQL

- Engineered full-stack web applications by integrating responsive user interfaces with scalable backend services, leveraging modern frameworks like **Next.js**, **tRPC**, and **PostgreSQL**.
- Self-hosted infrastructure on a **Linux VPS** with a CI/CD script for reliable deployment and extensibility.

## TECHNICAL SKILLS

**Languages:** Python, C, C++, C#, Java, HTML, CSS, JavaScript, TypeScript, SQL

**Libraries & Frameworks:** React, Next.js, Node.js, Vite, Tailwind CSS, tRPC, PostgreSQL, SQLite, MongoDB, Pandas, numpy, PyTorch, Electron, Google ADK

**Developer Tools:** Git, Visual Studio, PyCharm, Eclipse, Unity, Vercel, Prisma, Drizzle, Figma, Linux, Docker, npm